
Abstract Diagrams for the Working Mathematician

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Abstract

Within the niche of mathematicians, there's a niche of people who need to draw diagrams in abstract mathematics. Within *that* niche there's a niche of people who have experience in programming and feel comfortable writing code to produce graphics. This article is for them. It's a survey of the available tools and a showcase of a library I wrote specifically to simplify the process of drawing diagrams in differential geometry, topology, set theory, etc.

1 Introduction

Back in high school, I was attending a seminar on differential geometry, and I wanted to practice my $\text{\LaTeX}$ skills by typesetting the lecture notes. I found myself needing to draw simple surfaces such as those seen on Figure 1.

At that point I realized I wasn't sure how to draw these surfaces. First of all, I had to choose a tool, of which there are several:

- **Inkscape** (<https://www.inkscape.org>)

This is a popular and versatile GUI tool for drawing vectorized graphics, suitable when little precision is needed. In our case, however,

References

- [1] D.E. Knuth. *Seminumerical Algorithms*. Addison-Wesley, 1981.

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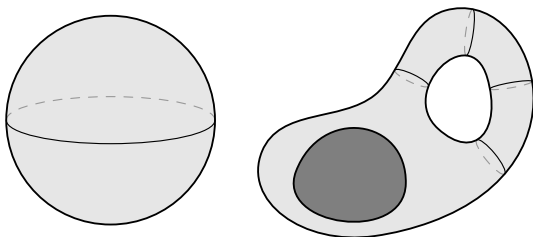


Figure 1: Basic two-dimensional manifolds